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Soccermatics Pro
Twelve

Project 2

Expected Danger Model

By Tiago Monteiro



Executive Summary

The Expected Danger (ED) model is a machine learning framework that quantifies how likely a pass will lead to a shot and goal. Unlike traditional assist or pass completion metrics, ED evaluates the **offensive value** of passes by combining spatial features, danger zone indicators, and progression dynamics.

Project Objective

The goal was to build a predictive model that answers: "How dangerous is this pass?" and pretends to enable clubs to:

- Identify players who consistently create high-value passing opportunities
- Understand spatial patterns in team attacking play
- Provide coaches with actionable insights beyond assists and key passes

Expected Danger (ED) Model Report

Model Architecture and Methodology

The Expected Danger (ED) calculation relies on a multiplicative two-stage pipeline designed to decompose the probability of a goal into distinct creation and conversion phases. The fundamental logic calculates ED by multiplying the probability of a shot occurring given a pass by the probability of a goal occurring given that shot. The first stage utilizes a Logistic Regression algorithm to predict shot probability, achieving a strong discrimination metric with a ROC-AUC of 0.7639. The second stage employs a Linear Regression model to estimate the goal probability for those shots, resulting in an R² of 0.0093. This low variance explanation in the second stage is an expected outcome given the inherent noise in goal conversion, which depends heavily on unmeasured variables such as goalkeeper positioning and shot technique.

Feature Engineering

The model input consists of 17 engineered features derived from raw pass events, which are categorized into spatial and enhanced metrics. The foundational spatial features capture the geometry of the play, including the

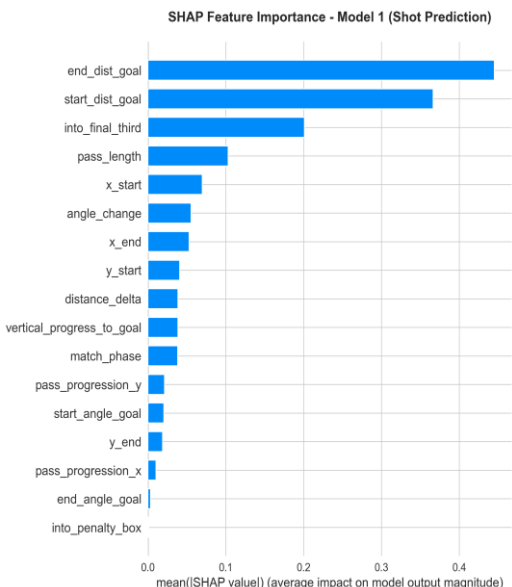
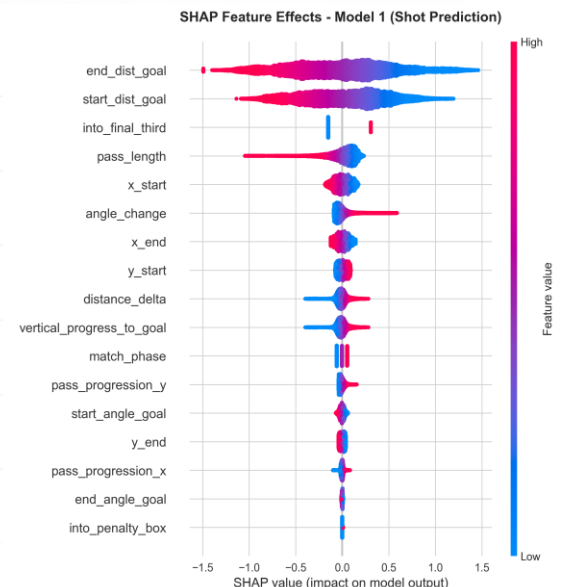
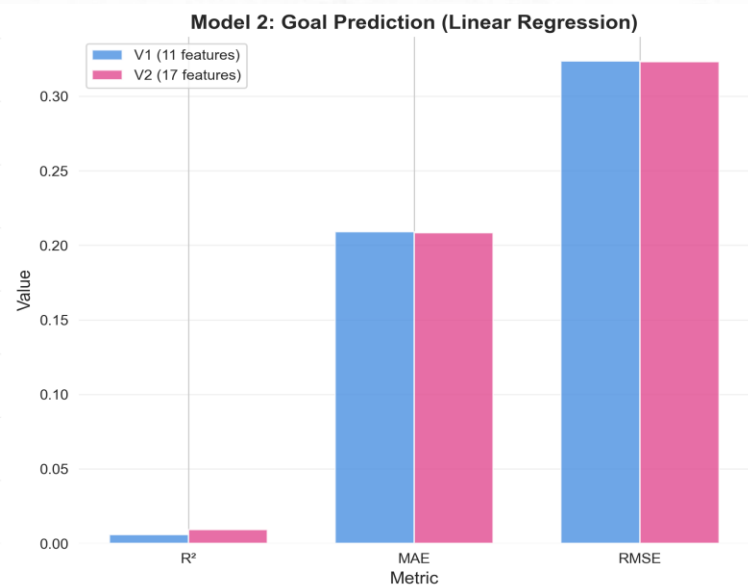
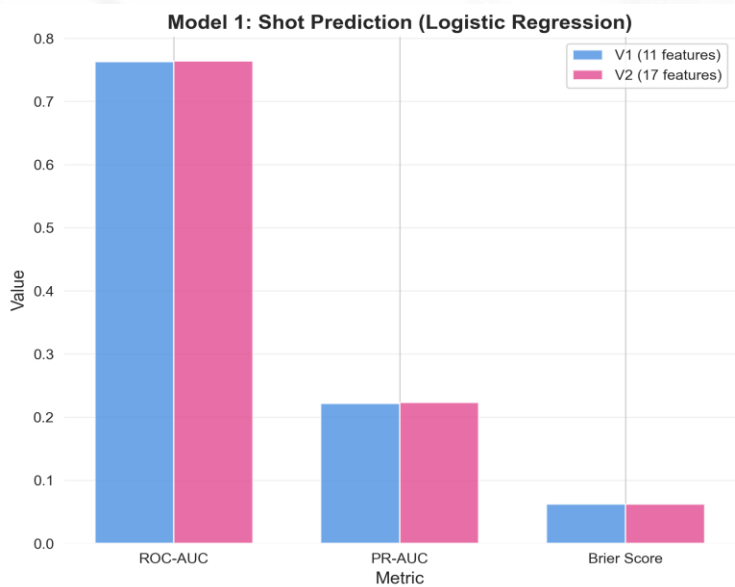
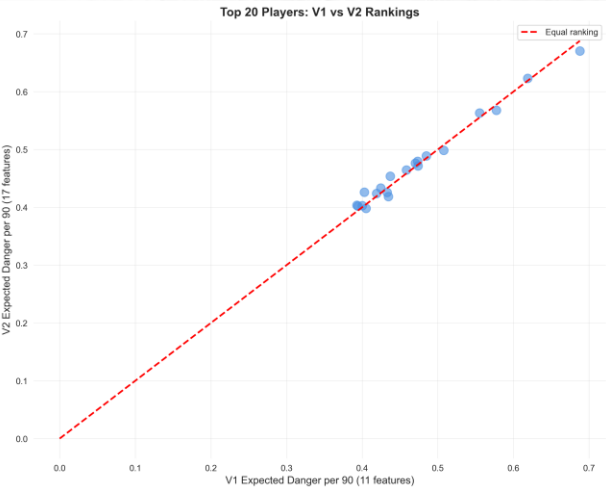
coordinates, distances, and shooting angles relative to the goal for both the origin and destination of the pass. Version 2 of the model expanded upon this by introducing eight enhanced features designed to capture context and momentum. These additions include directional progression vectors, the total length of the pass, binary indicators for entering the penalty box or the final third, and the specific match phase in which the event occurred.

Performance Analysis

A comparative analysis between Version 1 and Version 2 reveals that the addition of the eight enhanced features yielded only marginal performance gains. While the ROC-AUC improved slightly to 0.7639 and the Precision-Recall AUC rose to 0.2234, these represent increases of less than one percent over the baseline spatial model. Despite the class imbalance where only 8 percent of passes lead to shots, the model performs approximately 2.8 times better than a random baseline and demonstrates strong calibration with a Brier Score of 0.0621. The minimal uplift suggests that the core spatial metrics, particularly the distance to the goal, already capture the vast majority of the predictive signal.

SHAP Feature Importance and Insights

The application of SHAP analysis provides transparent insight into the factors driving the model's predictions. The results conclusively identify the destination of the pass as the dominant variable, with the distance from the pass end-point to the goal holding the highest mean absolute SHAP value of 0.45. This factor significantly outweighs the starting location of the pass and creates a clear hierarchy where proximity to the goal is the primary determinant of danger. The binary indicator for entering the final third serves as the third most important feature, effectively doubling the danger value compared to midfield possession. These findings offer a quantitative basis for tactical instruction, confirming that delivery location is far more critical than the distance covered by the ball. The model indicates that a pass ending 20 meters from the goal generates roughly three times more expected danger than one ending 30 meters out, validating a coaching emphasis on maximizing receiver depth.



3D Danger Surface Analysis

The 3D surface visualizations provide a topographical representation of each team's Expected Danger concentration, where the height of the surface corresponds to the density of dangerous pass creation. Analysis of elite teams such as Manchester City, Arsenal, and Liverpool reveals a distinct pattern characterized by sharp, concentrated peaks within the penalty area and half-spaces. Manchester City's surface is defined by the highest central peaks, indicating that their positional play successfully funnels danger through Zone 14, making almost every pass in that sector intentional and high-value. Arsenal displays a notable asymmetry with a strong peak in the left half-space, driven by the Ødegaard and Saka axis, while Liverpool presents the most difficult profile to defend, featuring dual peaks on both flanks and the center, reflecting a diverse creative approach that utilizes the full width of their 4-3-3 system. For opposing coaches, these topologies suggest a defensive strategy focused on blocking central lanes and compacting the shape in Zones 13 and 14, effectively accepting wide play where conversion rates are lower.

In contrast to the elite cohort, mid-table sides like Newcastle United and Aston Villa exhibit broader, flatter surfaces with only moderate elevations. Newcastle's data visualizes a rising plateau toward the box rather than a defined peak, suggesting they create danger progressively but lack precision in the final third, whereas Aston Villa shows a predictable left-sided bias. Lower-table teams, including Southampton and Ipswich Town, are characterized by flat surfaces with almost no significant elevation in the box.



Player Rankings and Tactical Analysis

The analysis of player rankings utilizes Expected Danger per 90 minutes (ED/90) as the primary efficiency metric, allowing for fair comparisons between players regardless of total minutes played. This normalization highlights the creative intensity of specific individuals rather than simply rewarding longevity. Leading the cohort is Trent Alexander-Arnold, who records an exceptional 0.67 ED/90. This figure is particularly notable for a defender and underscores his unique tactical role where Liverpool's setup maximizes his ability to deliver diagonal switches and progressive passes into the right half-space. Close behind is Arsenal's Martin Ødegaard at 0.62 ED/90, who distinguishes himself by combining this high efficiency with the highest volume of Total Expected Danger (16.14) in the dataset, a testament to his consistent operation in advanced playmaking zones.

Broader examination of the results uncovers significant global trends, most notably the dominance of Liverpool FC, who place five players in the top twenty rankings. This clustering suggests a systemic approach to chance creation that distributes responsibility across the backline and midfield. This aligns with the wider "Fullback Revolution" observed in the football, where seven of the top twenty creators are technically defenders. Modern fullbacks have effectively supplanted traditional wingers as primary creators in many systems. Conversely, the analysis identifies a "Forward Paradox" where elite attackers like Mohamed Salah record lower ED/90 figures (0.42). This lower ranking is structural rather than a performance deficit, as forwards typically function as the recipients of dangerous passes, converting Expected Danger into shots, rather than the generators of the passes themselves.

TOP DANGER CREATORS

Highest Expected Danger Generators | Premier League 23/24

#	PLAYER	CLUB	TOTAL ED	ED / 90
1	 T. Alexander-Arnold RB	 Liverpool	17.66	0.67
2	 M. Ødegaard RCMF3	 Arsenal	16.14	0.62
3	 K. De Bruyne AMF	 Manchester City	10.77	0.57
4	 K. Walker RB	 Manchester City	6.05	0.56
5	 Pedro Porro RB	 Tottenham Hotspur	14.48	0.50
6	 K. Trippier RB	 Newcastle United	7.13	0.49
7	 I. Maatsen LB	 Aston Villa	6.03	0.48
8	 M. Gusto RB	 Chelsea	9.81	0.48
9	 A. Robertson LB	 Liverpool	13.44	0.47
10	 A. Wharton RCMF	 Crystal Palace	6.81	0.46
11	 V. van Dijk LCB	 Liverpool	16.80	0.45
12	 C. Romero RCB	 Tottenham Hotspur	6.84	0.43
13	 W. Saliba RCB	 Arsenal	14.69	0.43
14	 M. Senesi LCB	 Bournemouth	5.23	0.43
15	 M. Kovačić DMF	 Manchester City	10.51	0.42
16	 Mohamed Salah RAMF	 Liverpool	15.73	0.42
17	 Gabriel Magalhaes LCB	 Arsenal	10.60	0.40
18	 A. Mac Allister LDMF	 Liverpool	11.67	0.40
19	 J. van Hecke RCB	 Brighton	13.31	0.40
20	 C. Palmer AMF	 Chelsea	14.15	0.40

Data: Twelve Soccermatics Pro • Premier League 2023-24

Position-Based
Rankings Analysis

An analysis of danger creation across different playing positions reveals distinct structural patterns in modern football. Defenders display a surprising capacity for high-value distribution, though Trent Alexander-Arnold remains a statistical outlier at 0.67 ED/90, operating far above the typical fullback range of 0.43 to 0.56. Interestingly, the data indicates that elite center-backs like William Saliba (0.43 ED/90) can now match the creative output of standard midfielders, reflecting the increasing responsibility placed on central defenders to initiate attacks. Midfielders exhibit the widest performance variance, ranging from 0.42 to 0.62 ED/90, with advanced playmakers such as Ødegaard and De Bruyne naturally leading the cohort, while deep-lying midfielders trail slightly behind. Forwards, by contrast, consistently record the lowest ED/90 figures (0.22–0.29), a finding that quantitatively reinforces their tactical role as the finishers, the recipients of dangerous passes, rather than the architects of the moves themselves.

The analysis further refines player evaluation through the "Rank Delta" metric, which measures the difference between a player's volume of dangerous passes and their per-90 efficiency ranking. A positive delta signifies a player whose efficiency rank outstrips their volume rank, indicating a style that prioritizes quality over quantity. Mateo Kovačić serves as the prime example of this efficiency, his substantial positive delta of +14 reveals that while he attempts fewer passes than other elite creators, the passes he does select possess exceptional danger value, making him one of the most efficient distributors in the dataset despite a lower overall volume.

POSITION RANKINGS

Expected Danger vs Danger Passes per 90



DEFENDERS

Player	Team	ED/90	DP/90	Rk Δ
 T. Alexander-Arnold RB		0.67	6.23	-
 K. Walker RB		0.56	5.50	+1
 Pedro Porro RB		0.50	4.51	+2
 K. Trippier RB		0.49	3.09	+9
 I. Maatsen LB		0.48	5.72	-3

MIDFIELDERS

Player	Team	ED/90	DP/90	Rk Δ
 M. Ødegaard RCMF3		0.62	5.79	-
 K. De Bruyne AMF		0.57	4.64	+1
 A. Wharton RCMF		0.46	4.02	+4
 M. Kovačić DMF		0.42	2.99	+14
 Mohamed Salah RAMF		0.42	4.34	-

FORWARDS

Player	Team	ED/90	DP/90	Rk Δ
 A. Isak CF		0.29	3.27	-
 Diogo Jota CF		0.24	1.96	+9
 Savinho RW		0.24	1.94	+9
 D. Núñez CF		0.23	3.04	-2
 J. Doku LW		0.22	2.60	-2

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